

SmartCRM AI

Product Requirements Document (PRD) — Release 3.0

Version 3.0 | Target: Product, Engineering, RevOps | Updated: October 2026

Executive Summary

SmartCRM AI is a next-generation customer relationship management platform engineered for modern B2B revenue teams, RevOps, and SDRs. Release 3.0 delivers native Google Maps Platform Lead Discovery & Interactive Spatial Intelligence, an overhauled 5-stage sales funnel with 1-click advancement, automated bulk file ingestion, and Google Gemini AI predictive scoring and sales copilot automation.

1. Product Vision & Strategic Objectives

Traditional CRMs function as passive administrative recording tools. SmartCRM AI transforms CRM software into an active revenue acceleration engine through geospatial prospect intelligence, transparent pipeline progression, and predictive AI automation.

1.1 Core Value Drivers

- Google Maps Geospatial Prospecting:** Discover verified local businesses via modern Google Places API (New) with interactive maps, rich InfoWindows, and 1-click CRM conversion.
- Transparent Sales Funnel:** Clear 5-step sales funnel (New Inbound -> Contacted -> Qualified -> Proposal Sent -> Closed Won) with column value aggregates and 1-click card advancement.
- Automated Lead Intelligence:** Quantitatively score leads on a 0-100 scale using predictive qualification signals with transparent rationales and next steps.
- High-Throughput File Ingestion:** Ingest up to 10MB of CSV/Excel files with intelligent header auto-mapping and real-time duplicate suppression.
- Zero Administrative Overhead:** Generate multi-tone customized outreach and executive meeting summaries with sub-second AI latency.

1.2 Success Metrics & KPIs

KPI / Metric	Target Baseline	Target Objective (v3.0)	Measurement Method
Rep Daily CRM Time	45 mins/day manual entry	< 15 mins/day	Session duration analytics
Prospect Discovery to CRM	20 mins per 10 prospects	< 10 seconds (1-click / bulk)	Google Maps lead logs
Lead Qualification Speed	4-8 hours post-inquiry	< 5 minutes	AI automated scoring latency
Outreach Drafting Latency	15 mins/email	< 30 seconds	Rep email generation logs
Pipeline Stage Clarity	Stale leads in vague stages	> 92% active stage accountability	Stage transition audit logs

2. Target User Personas & Workflows

2.1 Field Sales & Territory Account Executives (AEs)

- Pain Points:** Discovering target territory prospects, evaluating company credibility, managing late-stage deal momentum.

- **Core Workflows:** Searching Google Maps in target cities, evaluating ratings and websites on the interactive map, converting prospects with 1-click, and moving deals across the 5-stage funnel.

2.2 Sales Development Representatives (SDRs)

- **Pain Points:** Manual prospect sourcing, repetitive cold email drafting, and tedious spreadsheet data entry.
- **Core Workflows:** Using Google Maps niche presets (Restaurants, Tech, Real Estate) to bulk-import 20+ leads, running predictive AI Fit Scoring, and generating personalized outbound outreach.

2.3 VP of Sales & Revenue Operations (RevOps)

- **Pain Points:** Pipeline forecast inaccuracies, duplicate lead records, and lack of visibility into sales stage velocity.
- **Core Workflows:** Monitoring total pipeline value across stages in the interactive funnel bar, tracking conversion drop-offs, and auditing lead sources.

3. System Architecture & Data Model

SmartCRM AI employs a modern reactive full-stack architecture engineered for low-latency spatial discovery, seamless offline operation, and optional cloud synchronization via Supabase PostgreSQL.

Layer / Subsystem	Technology Stack	Operational Responsibility
Frontend Client	React 19, TypeScript, Tailwind CSS v4	Single-page application, responsive Kanban, split-view spatial discovery
Google Maps Platform	@vis.gl/react-google-maps, Places API (New)	Official React SDK, <Map>, <AdvancedMarker>, <Pin>, <InfoWindows>
API & Backend Server	Express, Node.js, TSX Runtime	RESTful CRUD endpoints, secure Maps proxying, deduplication, batch import
AI Engine	Google Gemini 3.8 Flash SDK (@google/gemai)	Lead qualification scoring, email drafting, sales copilot assistant
Persistence Engine	Dual-mode: JSON store + Supabase PostgreSQL	Resilient local caching + enterprise RLS multi-tenancy

4. Detailed Functional Requirements

4.1 Google Maps Lead Discovery & Interactive Map View

- **Official SDK & Map Integration:** Built using @vis.gl/react-google-maps with APIProvider, Map (DEMO_MAP_ID), and AdvancedMarker elements.
- **Modern Places API (New):** Uses https://places.googleapis.com/v1/places:searchText with field masking for maximum performance and cost optimization.
- **Interactive Spatial Sync:** Selecting a card highlights the map marker with glowing pulsing ring; clicking markers pans the camera and opens rich InfoWindows.
- **Visual Marker Pins:** Unsaved businesses display indigo pins; saved CRM leads display emerald pins with verified checkmark icons.
- **Rich InfoWindows:** Displays business name, category, rating stars, address, phone (tel:), website, Maps directions link, and "+ Add to Pipeline" CTA.
- **Niches & Location Presets:** 1-click industry buttons (Restaurants, Tech, Real Estate, Law, Healthcare) and city presets (SF, Austin, NYC, Chicago, Miami).
- **Discovery Filters & Sorting:** Filter by Minimum Rating (4.0+, 4.5+), Has Phone, Has Website, Unsaved Only; sort by rating, reviews, or name.
- **3-Way View Layouts:** Responsive toggle between Split View (List + Map), Map Only (Fullscreen), and List Only.

- **Bulk Pipeline Ingestion:** "Add All (X) to Pipeline" action allowing batch conversion of verified Google Maps listings into CRM leads.
- **Duplicate Prevention:** Real-time cross-checks by Google Maps placeId and normalized company name prevent duplicate records.

4.2 Transparent Sales Pipeline & Lead Management

- **Interactive Pipeline Stepper Funnel:** High-level visual funnel across 5 progressive stages: New Inbound -> Contacted -> Qualified -> Proposal Sent -> Closed Won.
- **Explicit Stage Definitions:** Each Kanban column header defines its qualification criteria with total deal counts and aggregate dollar values.
- **1-Click Stage Advancement:** Prominent forward action buttons on each card (Advance to Contacted "", Advance to Qualified "", Send Proposal "", Mark Won ☑️) and back-step arrow (+).
- **Google Maps Origin Badges:** Leads saved from Google Maps feature a dedicated badge, address preview, and a direct Google Maps navigation link.
- **Predictive AI Fit Scoring:** Prospects are scored 0-100 with color-coded badges (High Fit in green, Good Fit in amber, Low Fit in slate) with natural-language reasoning.
- **Smart Sales Actions:** Single-click triggers for AI Profile Dossier, AI Follow-Up Email Drafter, Lead Editing, and Deletion.

4.3 Automated Bulk Lead Import Engine (CSV / XLSX)

- **File Ingestion:** Support for CSV and Microsoft Excel (.xlsx, .xls) files up to 10MB via drag-and-drop or file selector.
- **Header Auto-Mapping:** Intelligent matching of spreadsheet columns to CRM fields with dropdown overrides.
- **Supported Lead Sources:** Strict normalization: Inbound Web, Outbound SDR, Referral, Partner Ecosystem, Conference, Google Maps.
- **Deduplication & Validation:** Email syntax checks, duplicate detection against database and file rows with optional auto-skip.
- **Live Preview & Audit Log:** Breakdown of Total Rows, Valid Records, Duplicates, and Invalids with persistent timestamped audit history.

4.4 Deal Pipeline & Revenue Forecasting

- **Deal Stages:** Discovery -> Demo Scheduled -> Proposal Sent -> Negotiation -> Closed Won -> Closed Lost.
- **Weighted Pipeline Calculation:** Probability-weighted forecasting based on stage conversion thresholds.
- **Stage Transitions:** Automated activity logging and timestamp updates upon any stage modification.

4.5 AI Sales Copilot (Conversational Assistant)

- **Pipeline Context Grounding:** Contextual grounding injecting current user pipeline metrics directly into Gemini prompts.
- **Capabilities:** Answering pipeline queries, identifying at-risk deals, drafting bespoke sales proposals, and coaching sales tactics.

5. Security, Data Privacy & Governance

SmartCRM AI adheres strictly to modern enterprise data governance and API security principles:

- **Zero Client-Side Secret Exposure:** Google Maps Platform API keys and Gemini API credentials reside securely on the backend server proxy.
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- **Multi-Tenant Data Isolation:** Supabase Row Level Security (RLS) policies enforce strict per-user boundaries (auth.uid() = user_id).
- **Local Fallback Encryption:** Offline browser storage isolates credentials and customer data per origin without external leak vectors.
- **Input Sanitization & Validation:** Strict file-type whitelisting, payload size limits, and regex validation on all ingest endpoints.

6. Non-Functional Requirements (NFRs)

Category	Specification	Verification Method
Performance	Map init < 500ms; Places search < 800ms; UI interactions < 100ms	Lighthouse & browser performance traces
Availability	99.9% uptime with instant client-side offline mode	Automated healthcheck polling (/api/health)
Scalability	Support 10,000+ active contacts and 5,000+ leads per account	Stress-testing JSON & Postgres indexes
Compliance	Zero deprecated Google Maps APIs; compliant with GMP Terms	Internal usage attribution & audit checks

7. Version History & Roadmap

Version	Release Date	Milestone Capabilities
v1.0	July 2026	Initial CRM MVP: Contacts, Companies, basic Leads list, and Deal tracking
v2.0	August 2026	Gemini AI Integration: Lead scoring, email generator, and executive dashboard
v2.4	September 2026	Automated Bulk Lead Import Engine (CSV/Excel) with header mapping and deduplication
v3.0 (Current)	October 2026	Google Maps Platform Lead Discovery, Interactive Spatial Map View (@vis.gl/react-google-maps), and Overhauled 5-Stage Sales Funnel with 1-Click Advancement

